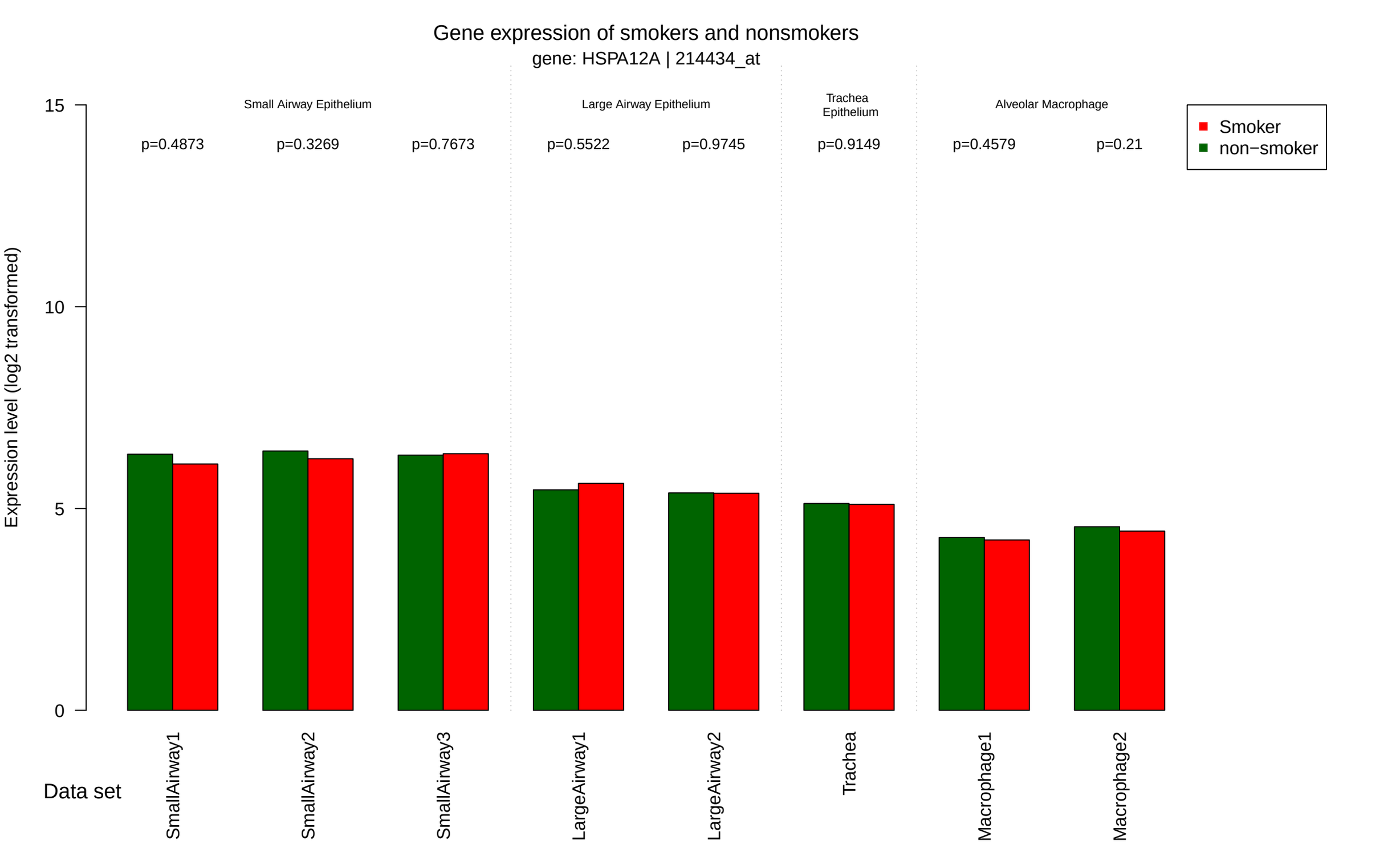
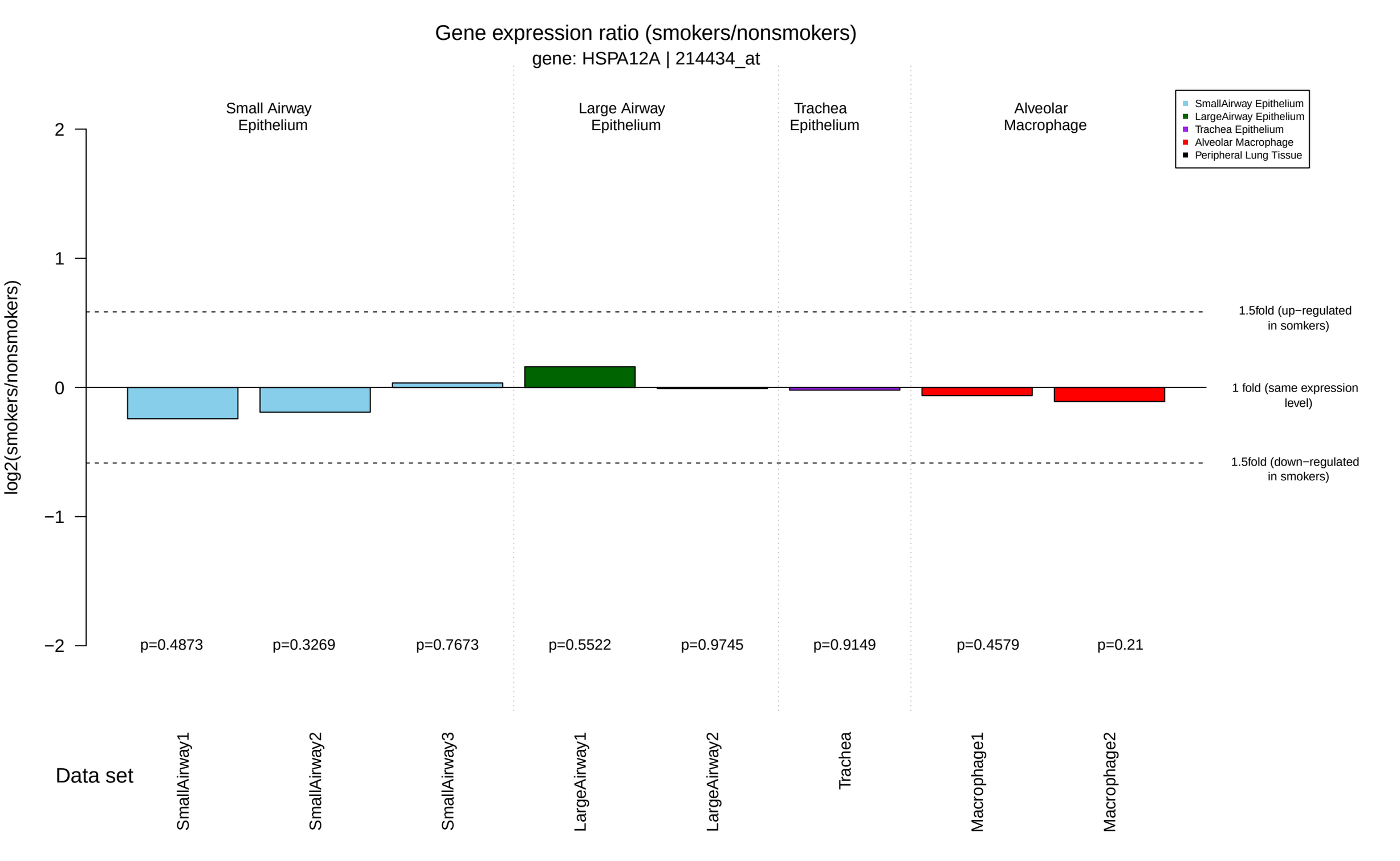
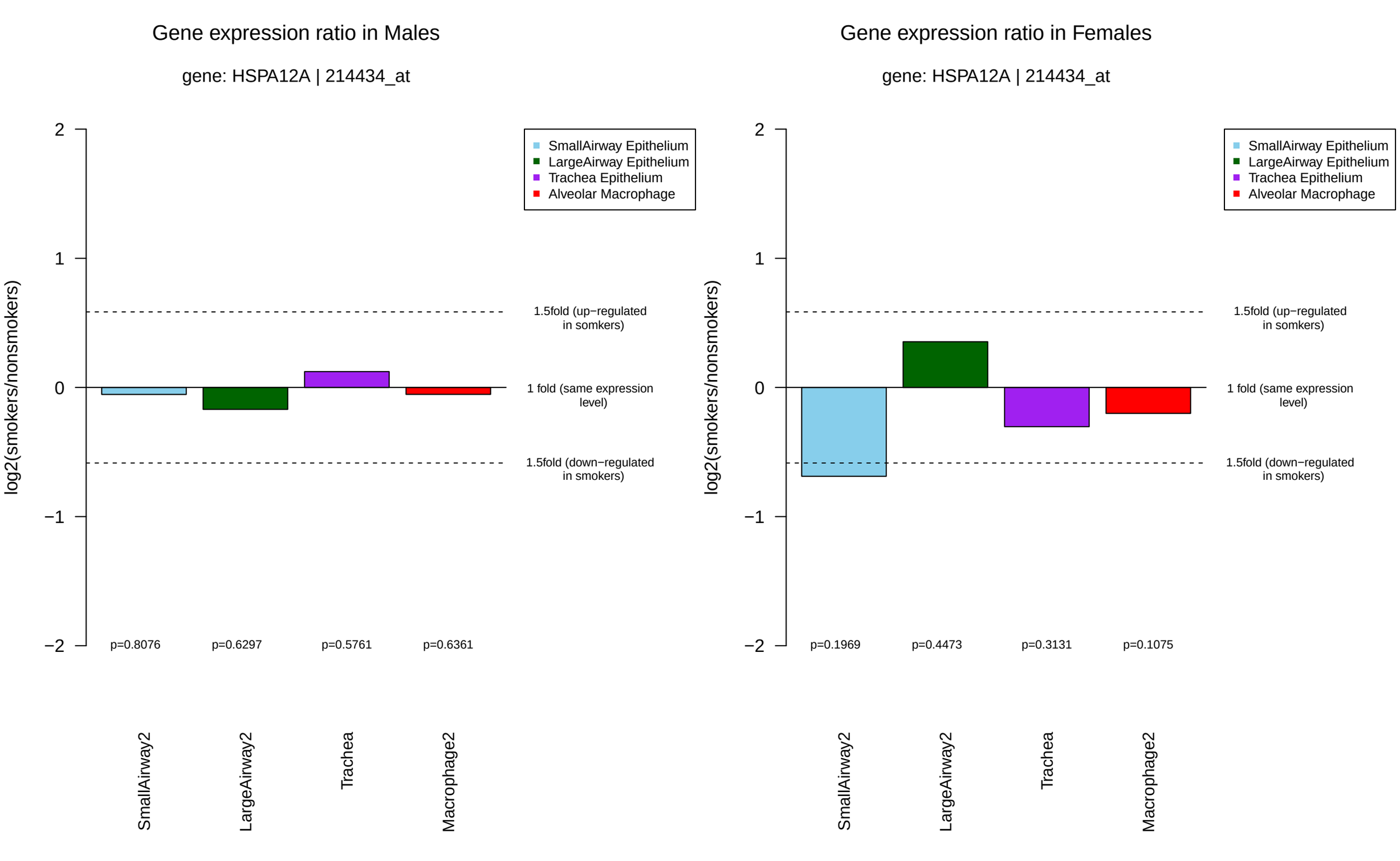
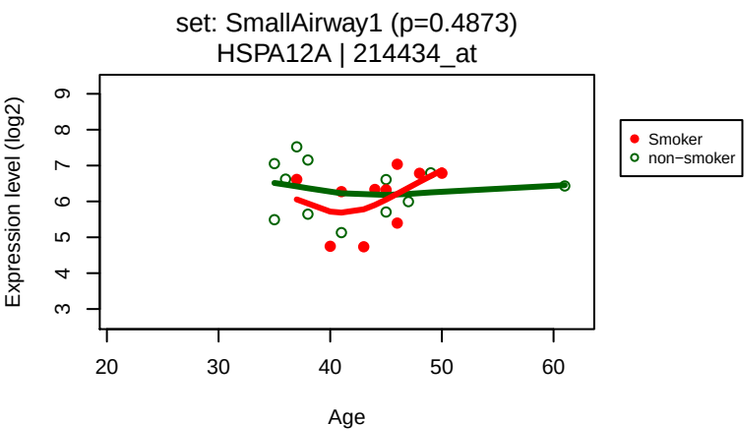
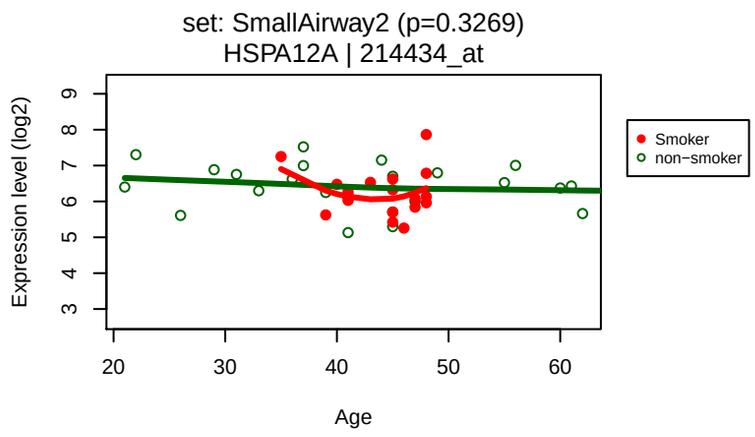
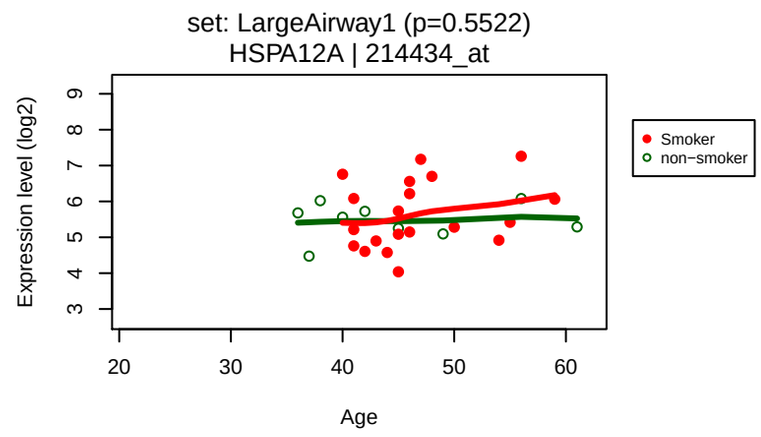

Gene expression of smokers and nonsmokers
gene: HSPA12A | 214434_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4873
p=0.3269
p=0.7673
p=0.5522
p=0.9745
p=0.9149
p=0.4579
p=0.21
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

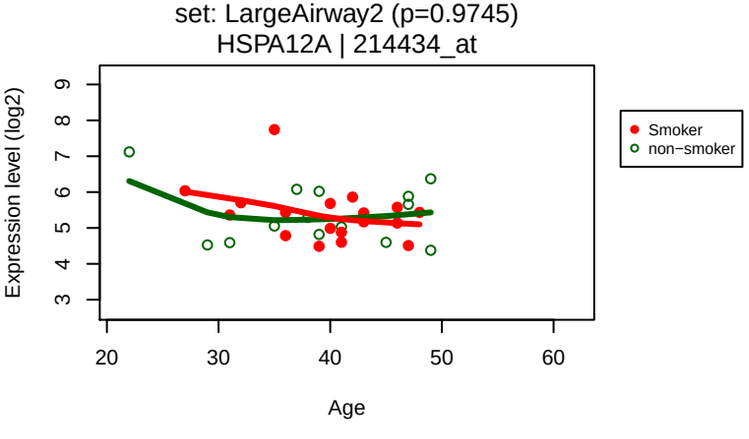

Gene expression ratio (smokers/nonsmokers)
gene: HSPA12A | 214434_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4873
p=0.3269
p=0.7673
p=0.5522
p=0.9745
p=0.9149
p=0.4579
p=0.21
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

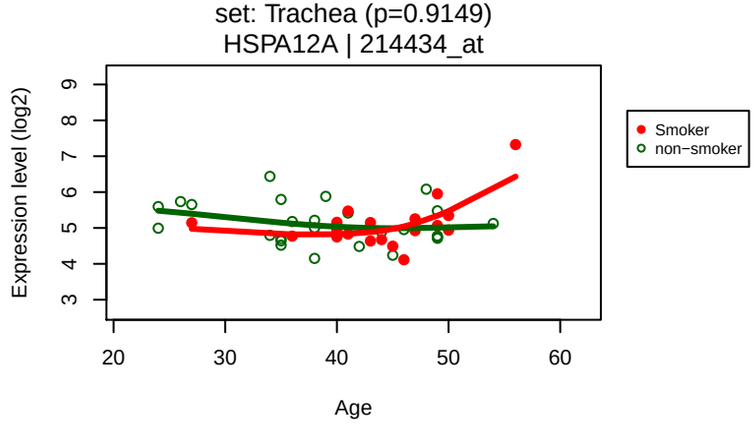

Gene expression ratio in Males
gene: HSPA12A | 214434_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8076
p=0.6297
p=0.5761
p=0.6361
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA12A | 214434_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1969
p=0.4473
p=0.3131
p=0.1075
SmallAirway2
LargeAirway2
Trachea
Macrophage2

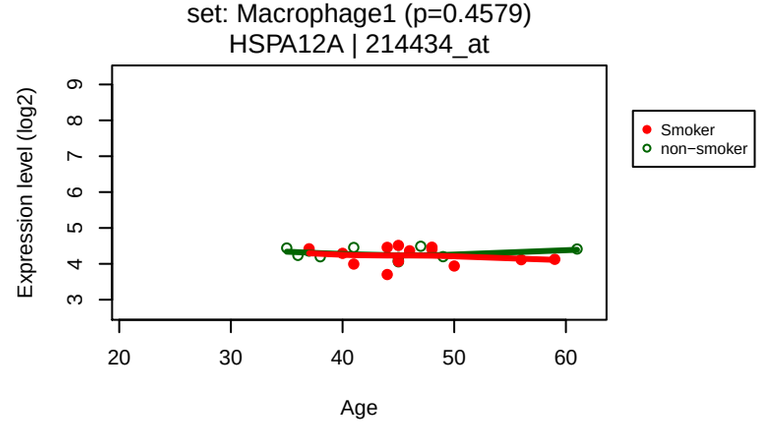

set: SmallAirway1 (p=0.4873)
HSPA12A | 214434_at
Expression level (log2)
Age
Smoker
non−smoker

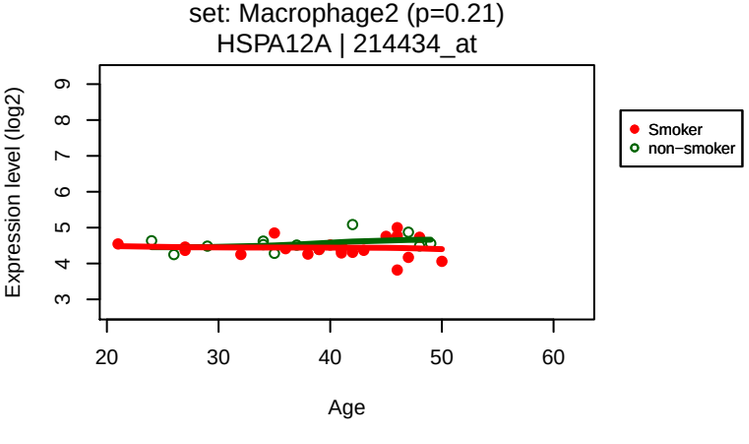

set: SmallAirway2 (p=0.3269)
HSPA12A | 214434_at
Expression level (log2)
Age
Smoker
non−smoker

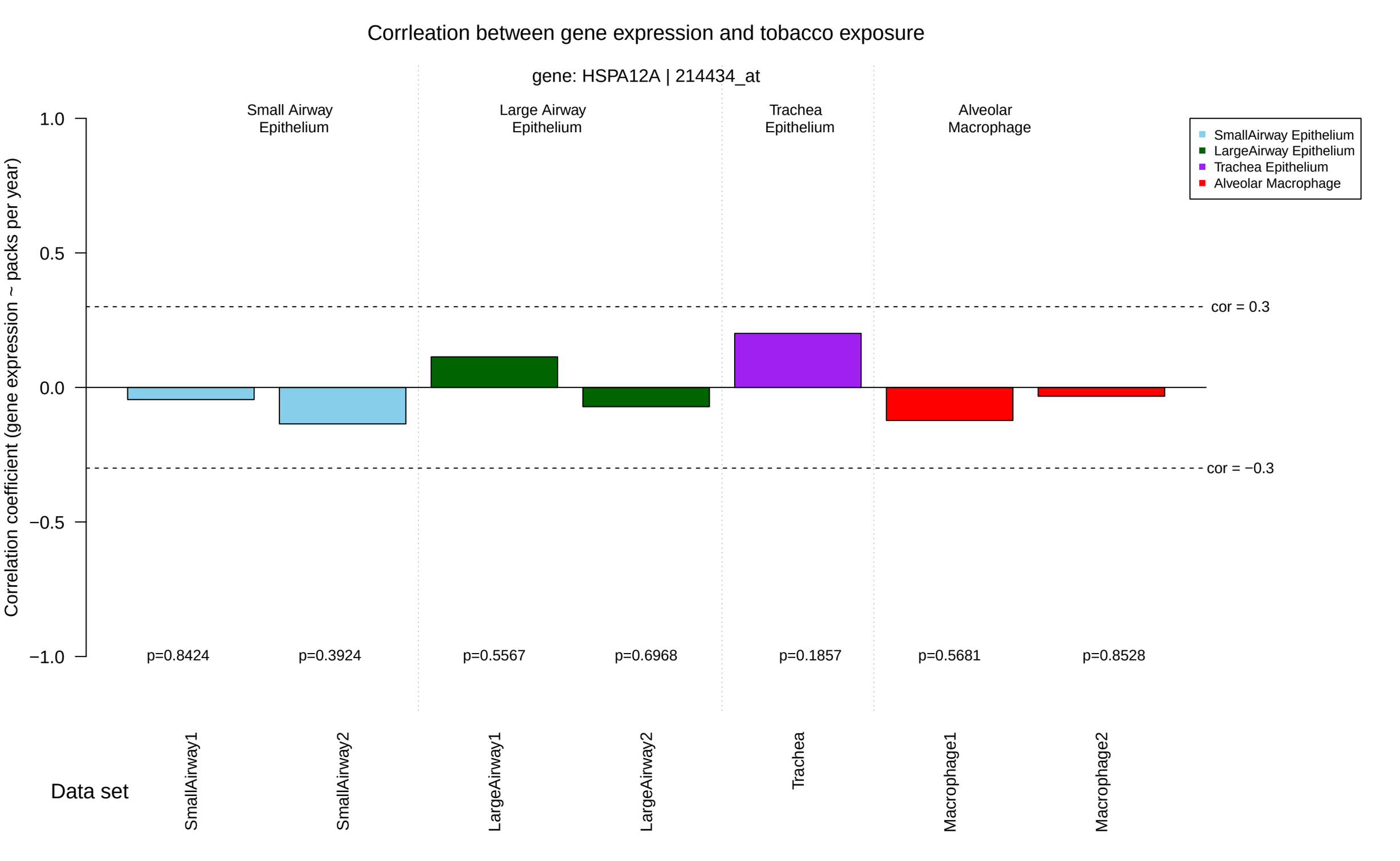

set: LargeAirway1 (p=0.5522)
HSPA12A | 214434_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9745)
HSPA12A | 214434_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9149)
HSPA12A | 214434_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4579)
HSPA12A | 214434_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.21)
HSPA12A | 214434_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA12A | 214434_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8424
p=0.3924
p=0.5567
p=0.6968
p=0.1857
p=0.5681
p=0.8528
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2